

Bill of Materials

All parts are through-hole, 2.54mm pitch.

Module	Part	Qty	Value	Note
All Modules	2.0mm Pitch 16pin Connector	1	Socket	Standard module connector
	Electrolytic Cap	1	10uf 16v	Main power decoupling
	Round Neodymium Magnets	4	5mm/5mm	
	PCB	1		
	Acrylic Face Plate	1		
	Screw	2	3M, 15mm	Telemetry needs 4
	Nut	4	3M	Telemetry needs 16
Abort	Circle Button	1	Red, 12mm	
	Safety Switch	1	Red	
	LED	1	Red, 10mm	
	Chip Socket	1	8 Pin	
	Resistor	1	1.2K, 1/4W	
	Resistor	1	10k, 1/4W	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny25	1		
Action Groups	Square Button	10	Blue	
	LED	10	Blue, 3mm	
	LED Socket	10	3mm	
	Chip Socket	2	8 Pin	
	Chip Socket	4	16 Pin	
	Resistor	10	10k, 1/4W	
	Resistor	10	62k, 1/4W	
	Ceramic Capacitor	6	100nf	Type 104
	Attiny45	2		
	74HC165	2		
	74HC595	2		
Analog	Square Button	2	Blue	
	Switch	2	SPST	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Chip Socket	1	16 Pin	
	Resistor	7	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny24	1		
	74HC165	1		
Analog Throttle	Square Button	2	Blue	
	Switch	1	SPST	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Chip Socket	1	16 Pin	
	Resistor	6	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny24	1		
	74HC165	1		
	Potentiometer	1	B20K	
	Potentiometer Cover Dial			
Camera	Square Button	4	Black	
	Square Button	6	White	
	Chip Socket	1	8 Pin	
	Chip Socket	2	16 Pin	
	Resistor	10	10k, 1/4W	
	Ceramic Capacitor	3	100nf	Type 104
	Attiny45	1		
	74HC165	2		

Module	Part	Qty	Value	Note
Control Systems	Square Button	10	White	
	Switch	2	SPST	
	LED	10	Orange, 3mm	
	LED Socket	10	3mm	
	Chip Socket	2	8 Pin	
	Chip Socket	4	16 Pin	
	Resistor	22	10k, 1/4W	
	Ceramic Capacitor	6	100nf	Type 104
	Attiny45	2		
	74HC165	2		
	74HC165	2		
Executive Actions (Control)	Square Button	1	White	
	Square Button	2	Blue	
	Circle Button	1	Green, 12mm	
	LED Button	1	Red, 16mm	
	Button Safety Cover	1	22x24mm	
	Switch	2	SPST	
	Key Switch	1	12mm	
	LED	3	Orange, 3mm	
	LED	2	Blue, 3mm	
	LED Socket	5	3mm	
	Chip Socket	2	8 Pin	
	Chip Socket	2	16 Pin	
	Resistor	1	1.2K, 1/4W	
	Resistor	10	10k, 1/4W	
	Resistor	2	62k, 1/4W	
	Ceramic Capacitor	4	100nf	Type 104
	Attiny45	2		
	74HC595	1		
	74HC165	1		
Executive Actions (Groups)	Square Button	5	Blue	
	Circle Button	1	Green, 12mm	
	LED Button	1	Red, 16mm	
	Button Safety Cover	1	22x24mm	
	Key Switch	1	12mm	
	LED	5	Blue, 3mm	
	LED Socket	5	3mm	
	Chip Socket	2	8 Pin	
	Chip Socket	2	16 Pin	
	Resistor	1	1.2K, 1/4W	
	Resistor	7	10k, 1/4W	
	Resistor	5	62k, 1/4W	
	Ceramic Capacitor	4	100nf	Type 104
	Attiny45	2		
	74HC595	1		
	74HC165	1		
Executive Actions	Circle Button	1	Green, 12mm	
	LED Button	1	Red, 16mm	
	Button Safety Cover	1	24x26mm	
	Key Switch	1	12mm	
	LED	1	Green, 5mm	
	LED Socket	1	5mm	
	Chip Socket	1	8 Pin	
	Resistor	1	300, 1/4W	
	Resistor	1	1.2K, 1/4W	
	Resistor	2	10k, 1/4W	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny25	1		

Module	Part	Qty	Value	Note
EVA	Square Button	6	White	
	Square Button	5	Green	
	LED Array	1	Green, 10 LED	
	Chip Socket	2	8 Pin	
	Chip Socket	4	16 Pin	
	Resistor	10	1.2K, 1/4W	
	Resistor	11	10k, 1/4W	
	Ceramic Capacitor	6	100nf	Type 104
	Attiny45	2		
	74HC1595	2		
	74HC165	2		
Navigation	Square Button	6	Black	
	D36 LED Switch	2	Yellow	
	Chip Socket	1	8 Pin	
	Chip Socket	1	16 Pin	
	Resistor	2	1.5k, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny45	1		
	74HC165	1		
Natigation (Time)	Square Button	2	Black	
	Square Button	2	Yellow	
	LED Button	1	Yellow, 16mm	
	D36 LED Switch	1	Yellow	
	6P Rocker Switch	1		
	Chip Socket	2	8 Pin	
	Chip Socket	1	16 Pin	
	Resistor	1	300, 1/4W	
	Resistor	1	1.5k, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	3	100nf	Type 104
	Attiny45	2		
	74HC165	1		
Rotation	Square Button	3	Green,	
	Switch	1	SPST	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Chip Socket	1	16 Pin	
	Resistor	6	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny24	1		
	74HC165	1		
Rotation (Throttle)	Square Button	2	Green,	
	Switch	1	SPST	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Chip Socket	1	16 Pin	
	Resistor	4	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny24	1		
	Potentiometer	1	B20K	
	Potentiometer Cover Dial	1		

Module	Part	Qty	Value	Note
Stage	Circle Button	1	Green, 12mm	
	Safety Switch	1	Green,	
	LED	1	Green, 10mm	
	Chip Socket	1	8 Pin	
	Resistor	1	150, 1/4W	
	Resistor	1	10k, 1/4W	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny25	1		
Telemetry	Square Button	4	Black	
	LCD Screen	1	128x64	
	Header Pins	1	1x20 Pin	Not socket
	Resistor	4	10k, 1/4W	
	Washer	8	3M	Depends on thickness
	Arduino Nano	1		
Throttle	Square Button	1	Yellow	
	Square Button	1	Red	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	Potentiometer	1	B20k, 6 Pin	Two parallel channels
	Throttle Lever	1		3D printed
	Throttle Friction Mechanism	1		3D printed
	Chip Socket	1	8 Pin	
	Resistor	2	10k, 1/4W	
	Diode	1	1N4148	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny25	1		
Time	Square Button	3	Black	
	Square Button	2	Yellow	
	LED Button	1	Yellow, 16mm	
	6P Rocker Switch	1		
	Chip Socket	2	8 Pin	
	Chip Socket	1	16 Pin	
	Resistor	1	300, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	3	100nf	Type 104
	Attiny45	2		
	74HC165	1		
Translation	Square Button	3	Green,	
	Switch	2	SPST	
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Resistor	5	10k, 1/4W	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny24	1		
Utility (Navigation)	Square Button	2	Black	
	Square Button	5	Green	
	D36 LED Switch	1	Yellow	
	LED Array	1	Green, 10 LED	
	Chip Socket	2	8 Pin	
	Chip Socket	3	16 Pin	
	Resistor	10	1.2K, 1/4W	
	Resistor	1	1.5k, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	5	100nf	Type 104
	Attiny45	2		
	74HC595	2		
	74HC165	1		

Module	Part	Qty	Value	Note
Utility (Time)	Square Button	3	Black	
	Square Button	5	Green	
	LED Array	1	Green, 10 LED	
	Chip Socket	2	8 Pin	
	Chip Socket	3	16 Pin	
	Resistor	10	1.2K, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	5	100nf	Type 104
	Attiny45	2		
	74HC595	2		
	74HC165	1		
Quad Hub	2.0mm Pitch 16pin Connector	4	Socket	Standard module connector
	Resistor	2	4.7k, 1/4W	
	Nut	6	3M	
	Arduino Micro	1		Can be substituted with a Nano
	Header Sockets	2	15 Pin, Socket	For the arduino to slot into
	Barrel Jack	1	2.1mm	
Octo Hub	2.0mm Pitch 16pin Connector	8	Socket	Standard module connector
	Resistor	2	4.7k, 1/4W	
	Nut	6	3M	
	Arduino Micro	1		Can be substituted with a Nano
	Header Sockets	2	15 Pin, Socket	For the arduino to slot into
	Barrel Jack	1	2.1mm	
Miscellaneous	Power Supply	1	5V	Center 5V, Sleeve ground
	Micro USB cable	1		
	Neodymium Magnets	10	5mm/5mm	
	Ribbon Cable	2-3m	16 wire, 1mm	Depending on size
	Ink	1	Black	Alcohol or acetone based
	Acrylic Container	1		Depending on size